FIREWALLS, NAT AND PACKET FILTERING

Course: Diploma in Cyber Security

Name: HET ACHARYA

Introduction

Firewalls and filtering methods help control which network traffic is allowed or blocked. NAT allows many private devices to share a public internet address. In this practical I worked with a Windows firewall rule and simple NAT and filtering examples.

Task 1: Block incoming port 8080

On Windows, I would open Windows Defender Firewall with Advanced Security, choose Inbound Rules, create a New Rule, select Port, choose TCP and enter 8080, select Block the connection, apply the required profiles, and give the rule a clear name such as Block Incoming 8080.

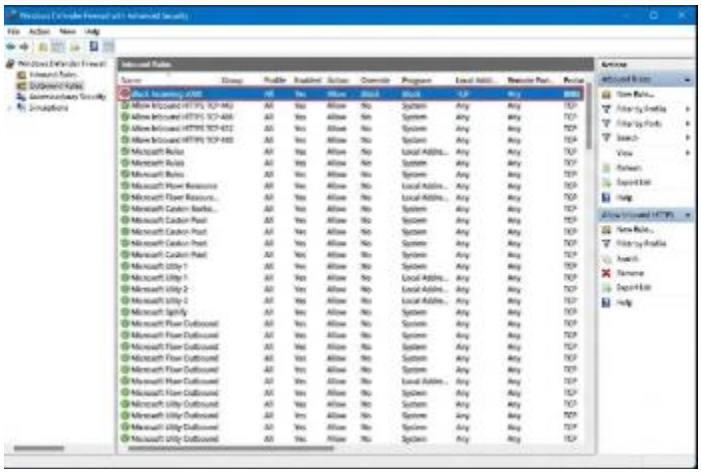


Figure: Windows Defender Firewall Inbound Rule Blocking Inbound TCP Port 8080

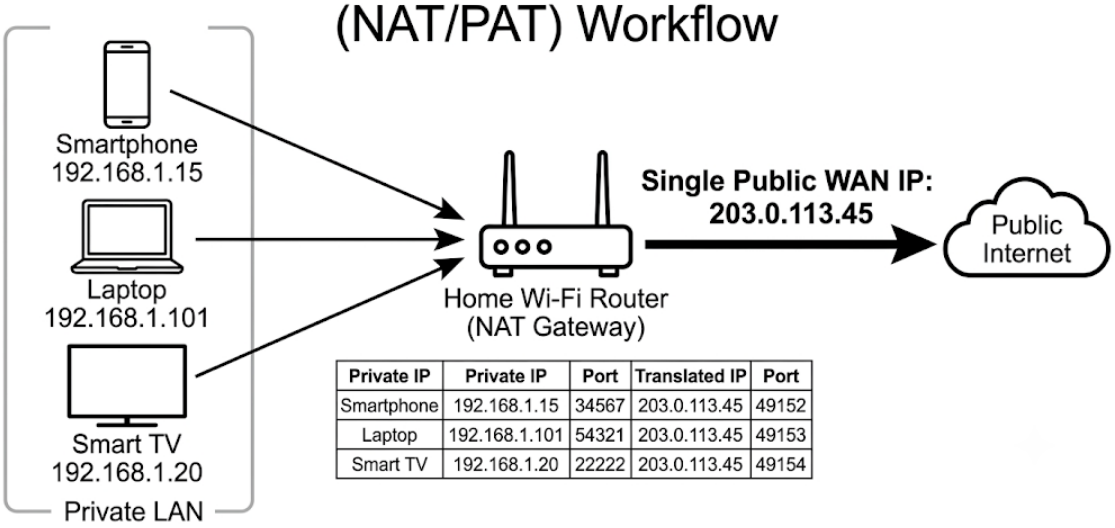
Configuration Summary:

- **Rule Type:** Inbound Port Filter.
- **Protocol & Port:** TCP / Local Port 8080 (commonly used for alternate web/proxy services).
- **Enforced Action:** Block the connection (Action: Block).
- **Applied Profiles:** Domain, Private, Public.
- **Security Objective:** Restricts external hosts from establishing incoming sessions to unauthorized web services, proxies, or development HTTP servers listening on port 8080.

Task 2: Explain NAT

A home router normally gives private IP addresses to devices such as a phone, laptop and smart TV. When these devices access the internet, the router translates their private addresses to the public address supplied by the ISP.

Figure: Figure: Network Address Translat) Workflow
(NAT/PAT) Workflow



Private IP	Private IP	Port	Translated IP	Port
Smartphone	192.168.1.15	34567	203.0.113.45	49152
Laptop	192.168.1.101	54321	203.0.113.45	49153
Smart TV	192.168.1.20	22222	203.0.113.45	49154

Figure: Network Address Translation (NAT / PAT) Process from Private LAN to Public Internet

Technical Explanation & Flow:

- **Private Addressing (Inside Local): Devices on the home network (Smartphone, Laptop, Smart TV) are leased private RFC 1918 addresses (192.168.1.0/24) by the router's local DHCP server. These addresses are non-routable on the global internet.**
- **NAT / PAT Translation (Inside Global): When an internal host initiates an outbound web session, the router modifies the source IP in the packet header, replacing the private address with the single public IPv4 address assigned by the ISP.**

- **Port Address Translation (PAT): The router tracks outgoing connections using distinct source port numbers in its NAT state table, ensuring return packets from the internet are routed back to the exact initiating internal device.**

Task 3: Port filtering on port 25

If the router supports outbound port filtering, I would create a rule to block TCP port 25. SMTP port 25 is commonly used for mail transfer between servers, and blocking direct outbound use can reduce some unwanted mail or spam activity from infected devices.

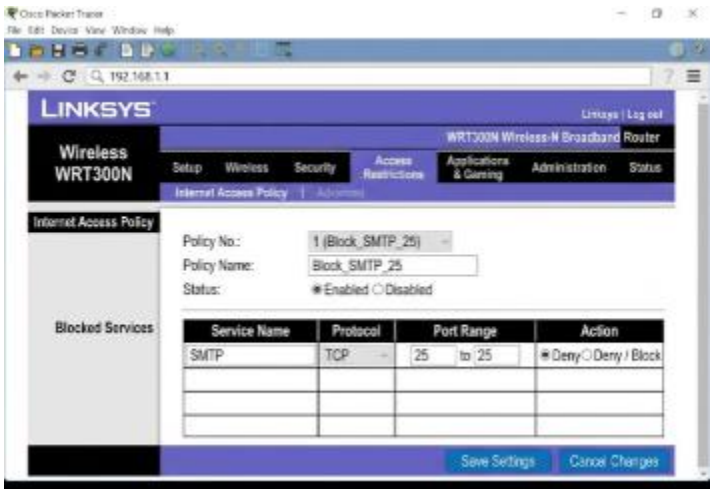


Figure: Wireless Router Access Restriction Rule Blocking Outbound TCP Port 25 (SMTP)

Task 4: Packet filtering

For each packet, I would compare the source IP, destination IP, protocol and port with the trainer's rule

set. A packet matching an allow rule is permitted, while a packet matching a block rule is denied.

Task 5: Firewall use by large apps

Large services can use firewalls and filtering at several points. Examples include blocking unwanted ports, limiting traffic from suspicious sources, and protecting server networks with rules that allow only required services.

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.